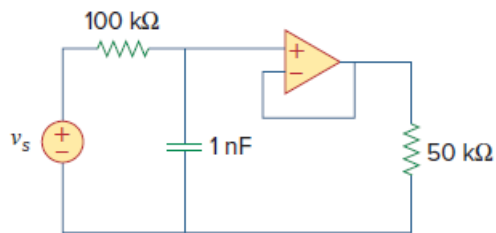


Problem

In the op amp circuit of Fig. 11.84, $v_s = 4 \cos 104t$ V. Find the average power delivered to the $50\text{-k}\Omega$ resistor.

Figure 11.84:



Step-by-step solution

Step 1 of 3

$$\omega = 10^4 \text{ rad/sec}$$

$$C = 1 \text{ nF}$$

$$1 \text{ nF} = \frac{1}{j\omega C}$$

$$1 \text{ nF} = \frac{-j}{10^4 \times 10^{-9}}$$

$$1 \text{ nF} = -j100 \text{ k}\Omega$$

Step 2 of 3

At the non inverting terminal,

$$\frac{4 \angle 0^\circ - V_0}{100} = \frac{V_0}{-j100}$$

$$V_0 = \frac{4}{1+j}$$

$$V_0 = \frac{4}{\sqrt{2}} \cos(10^4 t - 45^\circ)$$

Step 3 of 3

$$P_{50\Omega} = \frac{V_{rms}^2}{R_{50\Omega}}$$

$$P_{50\Omega} = \left(\frac{4}{\sqrt{2}} \frac{1}{\sqrt{2}} \right)^2 \left(\frac{1}{50 \times 10^3} \right)$$

$$\boxed{P_{50\Omega} = 80 \mu \text{ W}}$$